

# Computational Fluid Dynamics Modelling of Vacuum Cold Spray for In-Space Surface Repair

Ethan Liu

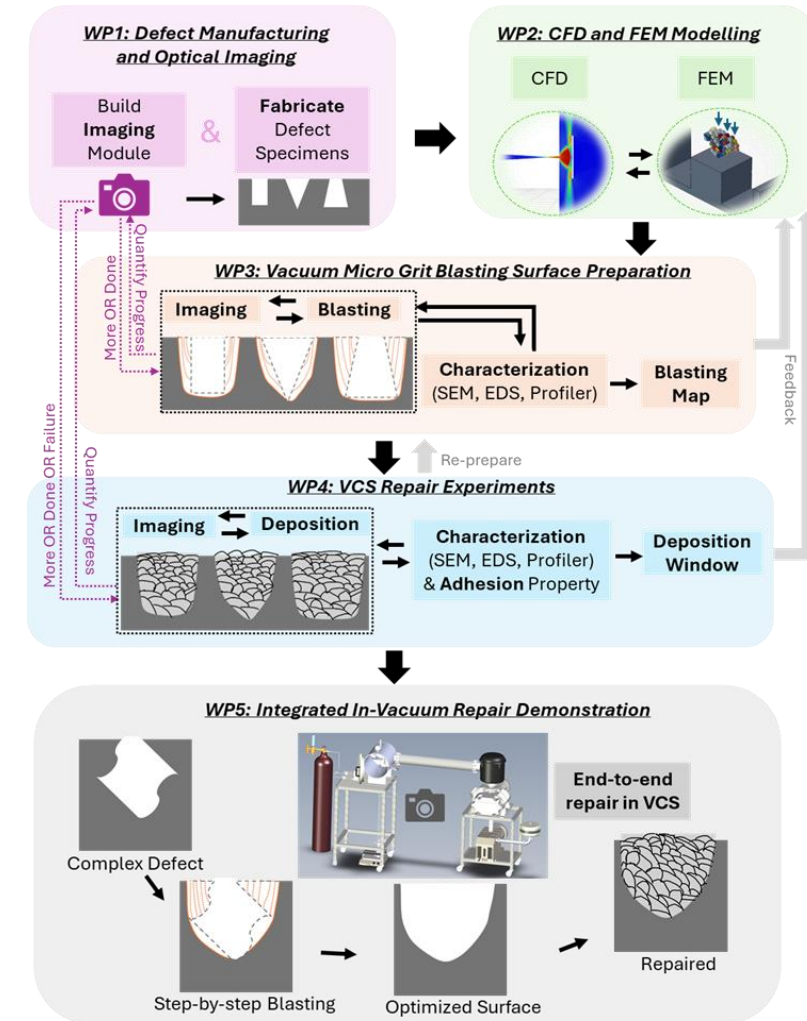
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Centre for Advanced Coating Technologies

# Why In-Space Repair?

- **Small defects can become major problems in space**
  - Spacecraft surfaces are exposed to debris impacts, wear, and thermal cycling.
  - Returning damaged components to Earth may be impractical.
  - Repair systems must operate locally and under vacuum.
  - Sub-millimetre defects require precise detection and material deposition.

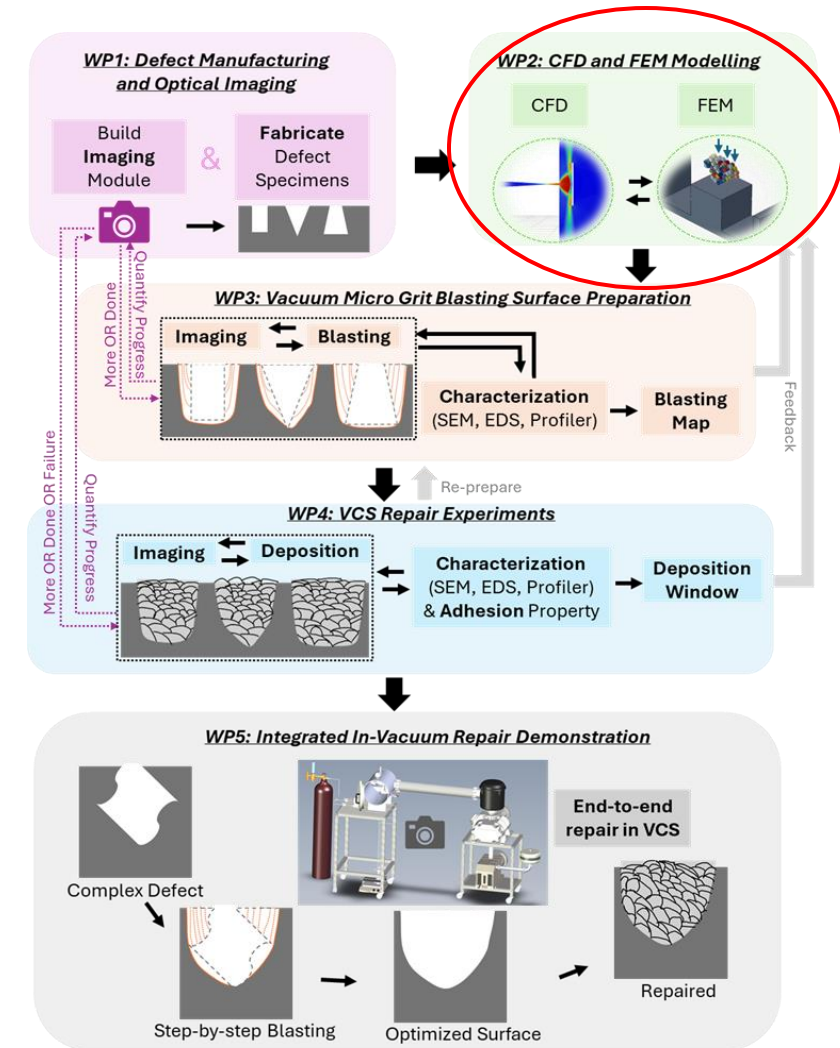


# My Role

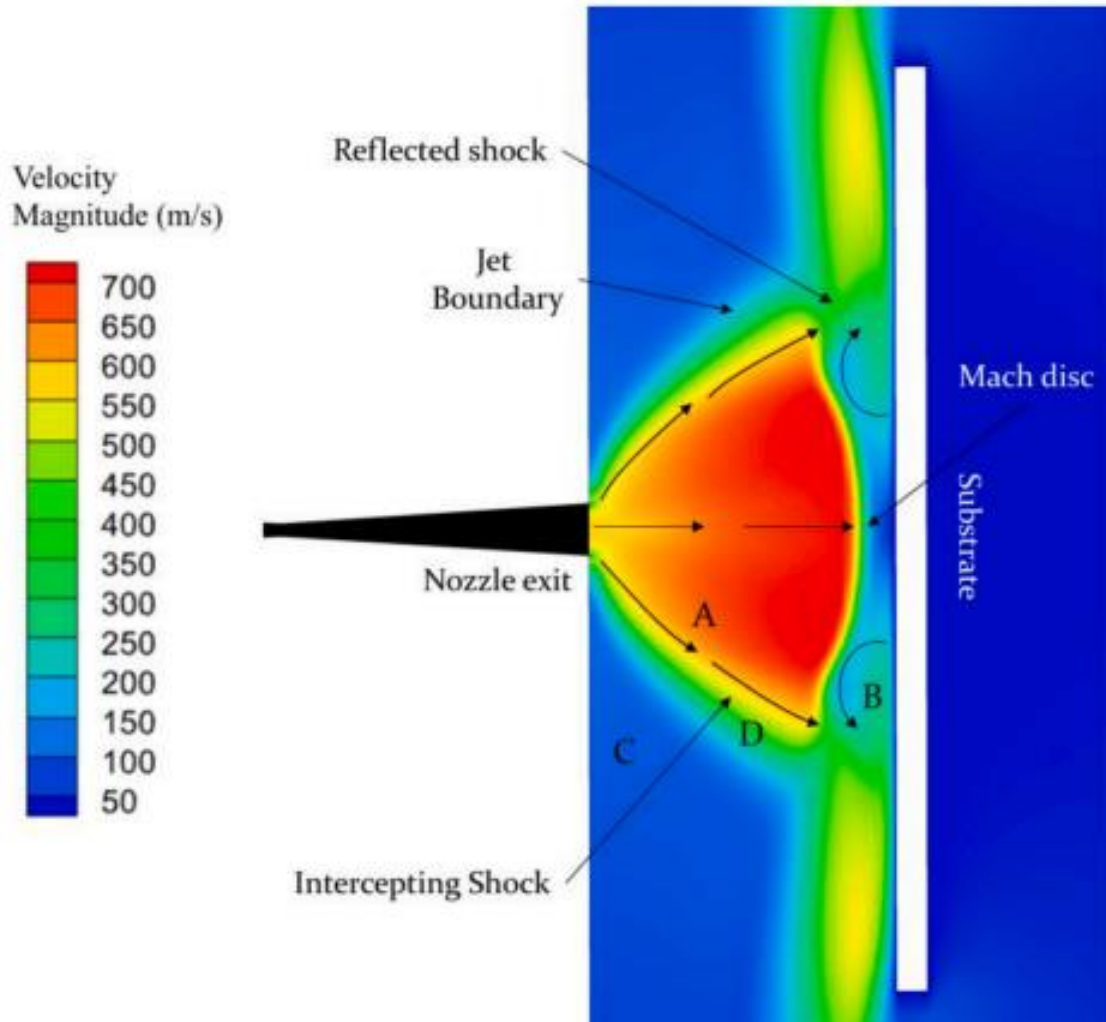
The goal of this summer was to **develop** and **validate** CFD models to predict particle , and deposition in a vacuum cold spray system.

## My Objectives:

- Refine CFD models in ANSYS Fluent
- Study gas flow and particle trajectories under vacuum
- Automate large parametric studies using PyANSYS and HPC
- Prepare simulation data for comparison with experiments

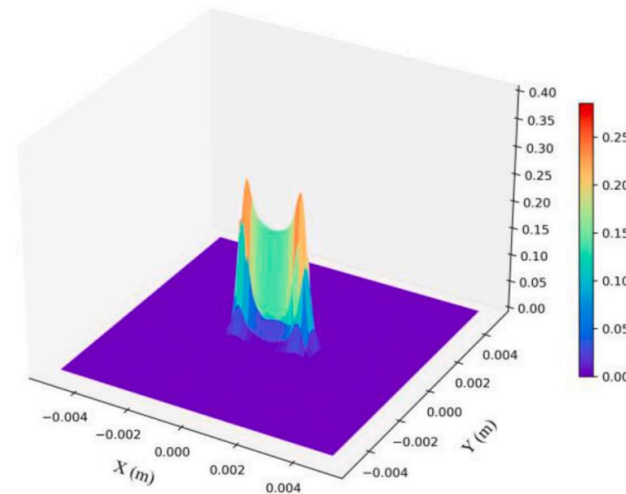


# Background – CFD

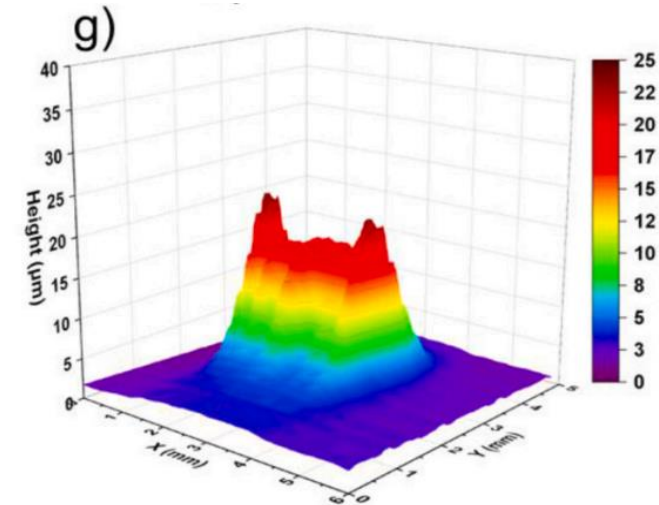


Gas flow of CD-slit Nozzle (Mahendhar, 2024)

- Experiments alone cannot efficiently test every operating condition.
- Modified CFD models can predict help predict deposition of future



Kernel Density Estimation from Discrete Phase Model (Mahendhar, 2024)



3D Surface Profile of Deposition (Mahendhar, 2024)

# Modelling narrows a large experimental design space

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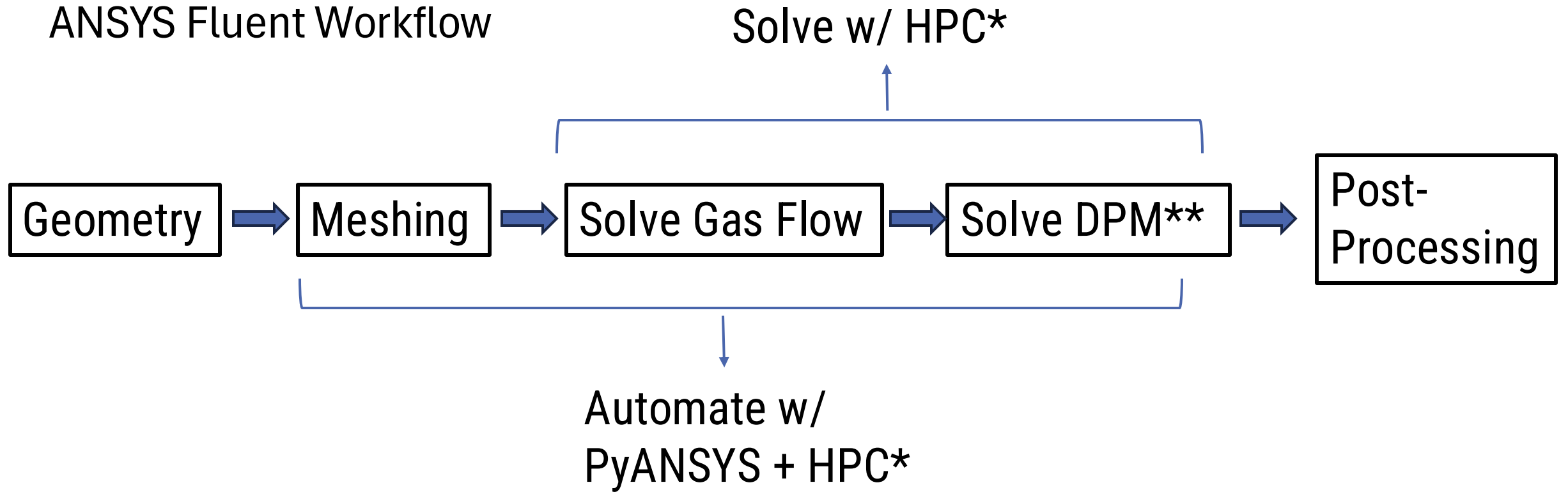
VCS Performance depends on interacting operating, geometric, and particle, parameters

| Operating Conditions                                        | Geometry                                                               | Particles                              |
|-------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------|
| <b>Gas Flow Rate</b><br>Chamber Pressure<br>Gas Temperature | <b>Nozzle Geometry</b><br><b>Stand-off distance</b><br>Defect Geometry | Diameter<br>Density<br><b>Material</b> |

**Simulations can "screen" combinations before time and materials are used to experiment**

# Initial Model Recreation + HPC

ANSYS Fluent Workflow



\*High Performance Computing  
\*\* Discrete Phase Model

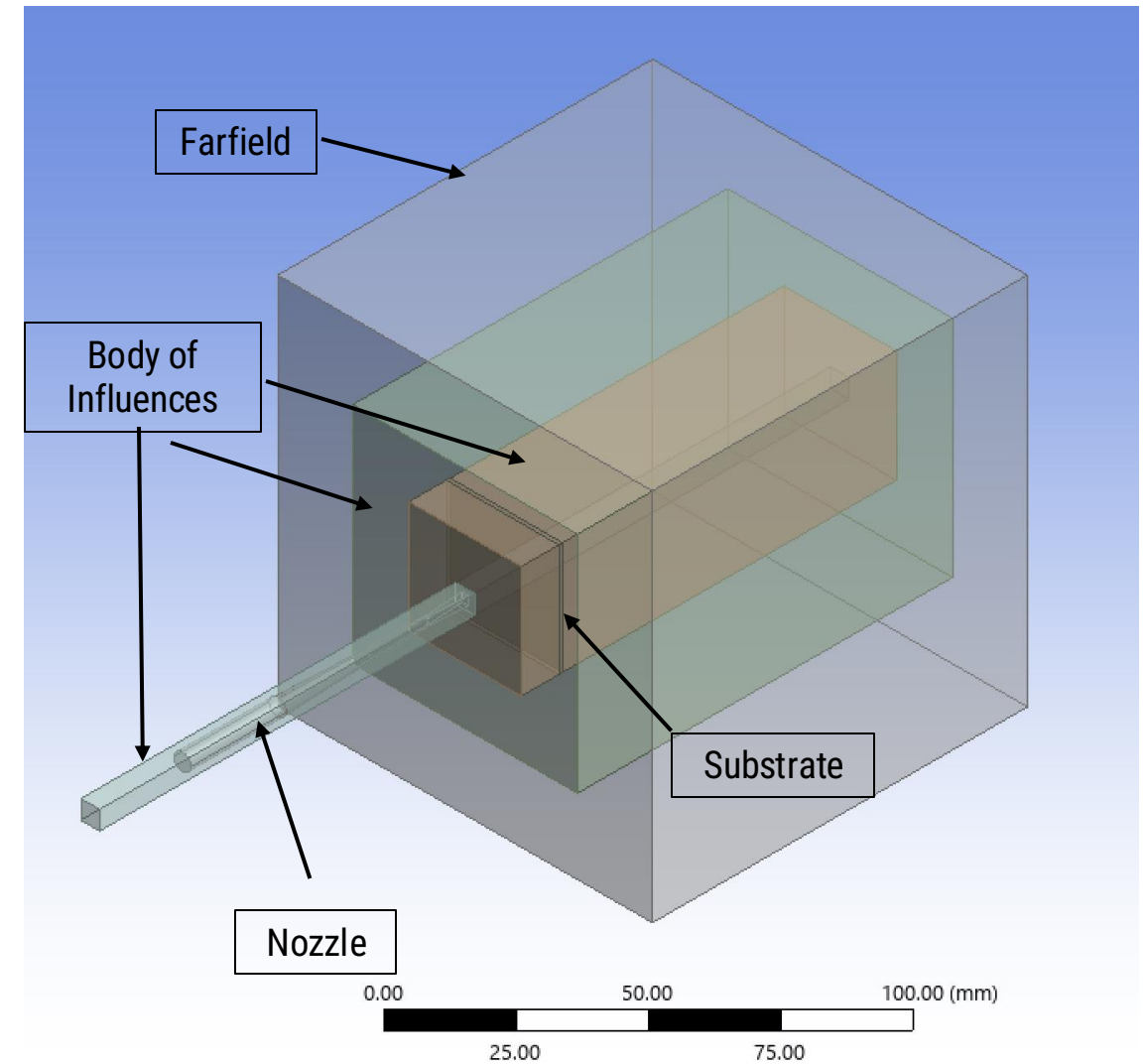
# CFD Model Setup

## Continuous Phase:

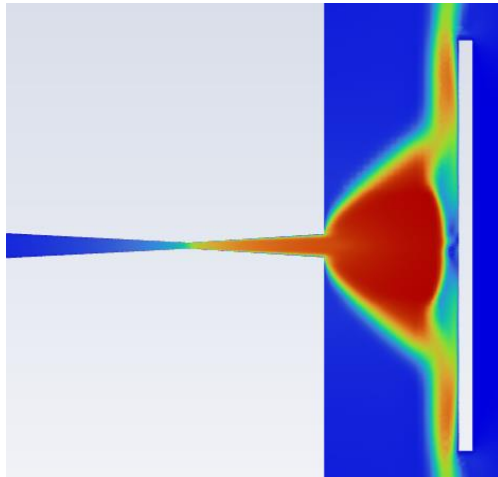
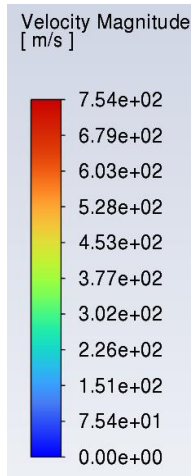
- Compressible N2
- Ideal gas
- SST k- $\omega$  turbulence

## Discrete Phase Model:

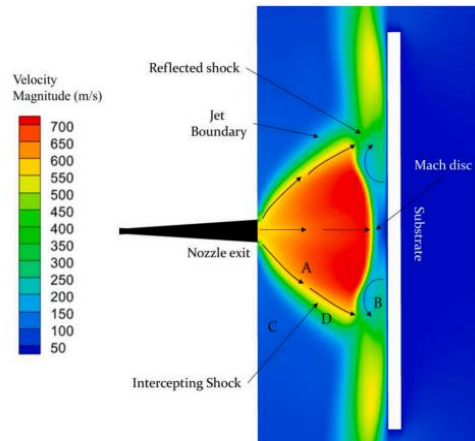
- Alumina or Copper
- Eulerian-Lagrangian approach
- Drag & Heat transfer
- Particle-gas interactions



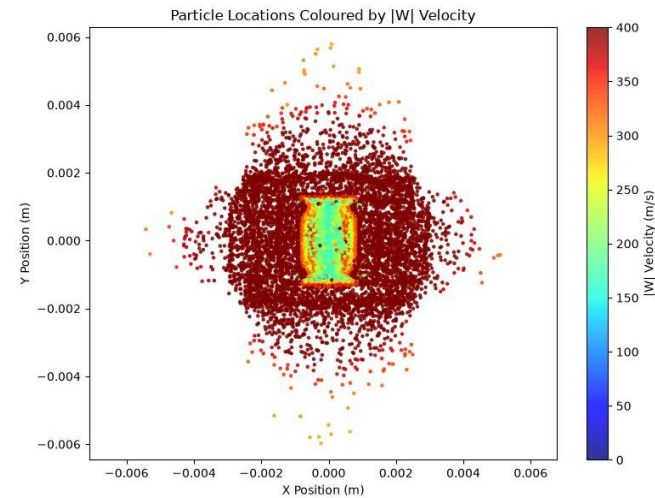
# Initial CFD Validation using CD-slit



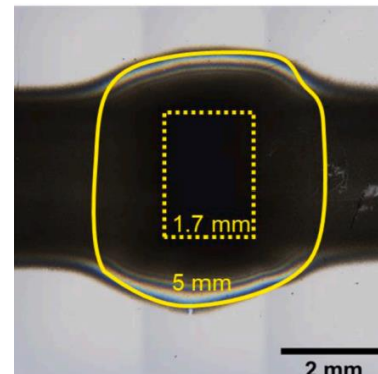
Gas flow contour



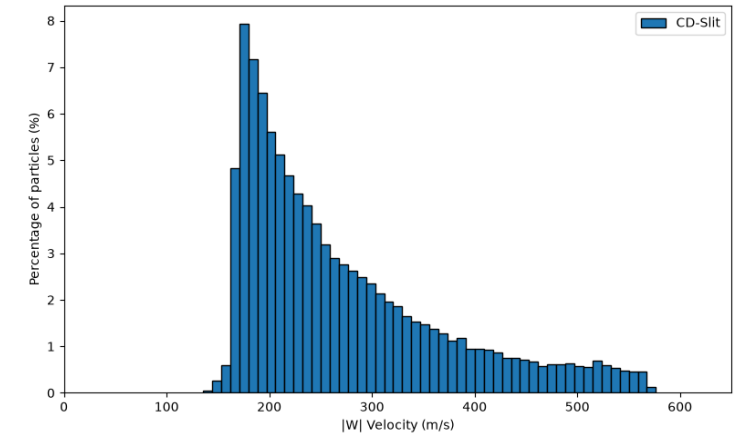
Gas flow of CD-slit Nozzle (Mahendhar, 2024)



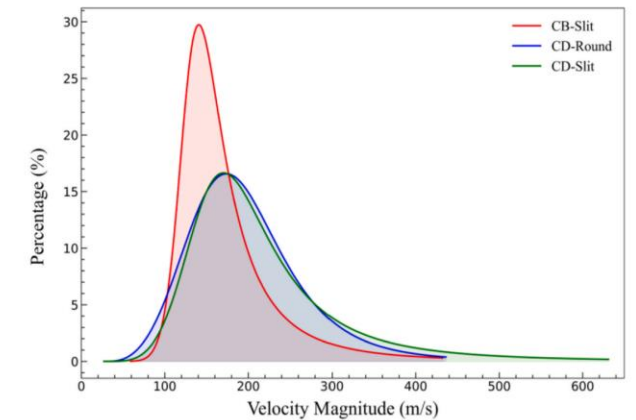
Particle Landing Scatter Diagram



Optical Micrograph of Deposition (Mahendhar, 2024)



Particle Impact Velocity Histogram



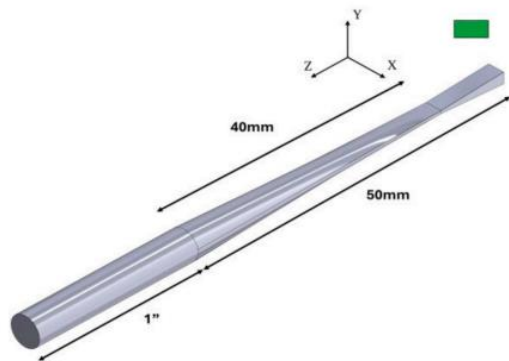
Particle Velocity Distribution (Mahendhar, 2024)

# Building a Foundational Database of Models

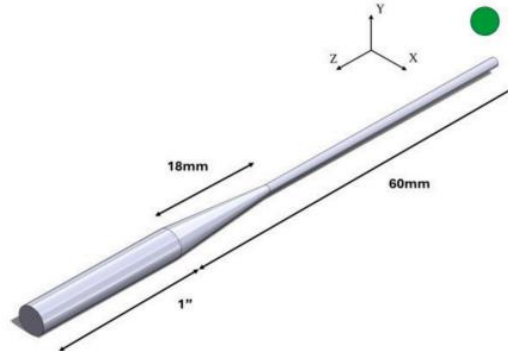
## PyANSYS Workflow:

- Update Case Parameters
- Solve w/ Fluent via HPC
- Monitor Convergence
- Track Failures and Timeouts
- Export Gas & Particle Data
- Continue through remaining cases

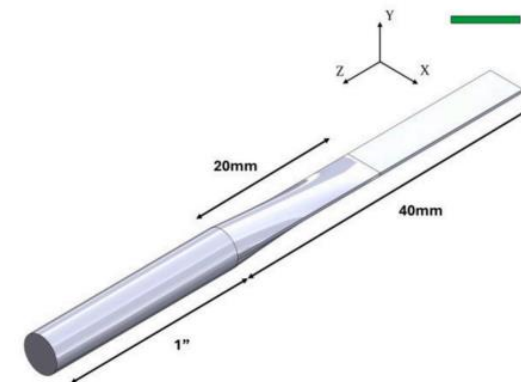
| Parameters         | Values                     |
|--------------------|----------------------------|
| Gas Flow Rate      | 3, 5, 10, 15 SLM           |
| Gas                | N2, He                     |
| Nozzle Geometry    | CD-slit, CB-slit, CD-round |
| Standoff Distance  | 3, 6, 10 mm                |
| Substrate Angle    | Flat, 30, 60 degrees       |
| Injection Particle | Al2O3, Cu                  |



(a) CD-Slit



(b) CD-Round

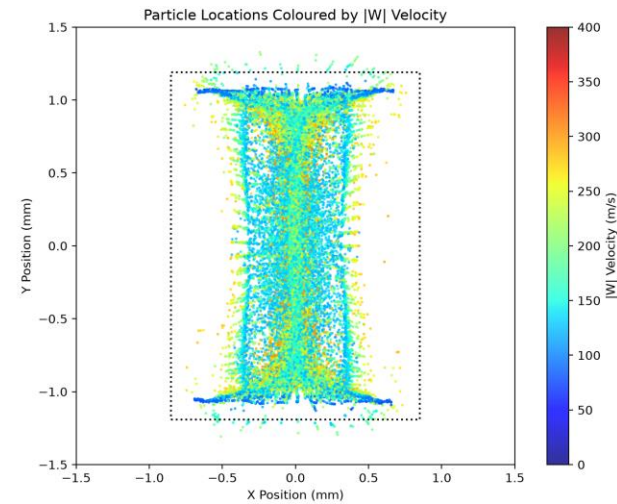


(c) CB-Slit

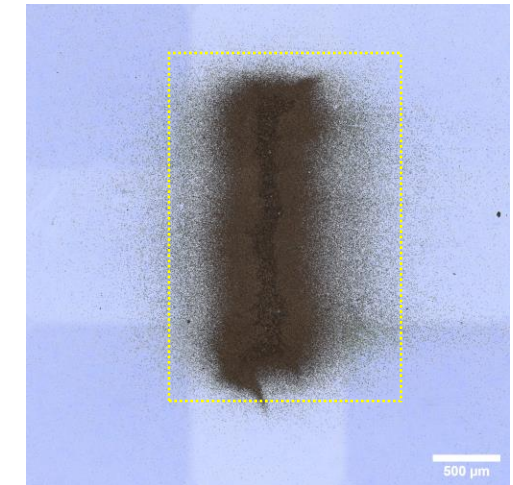
Nozzle Geometries (Mahendhar, 2024)

# Validating Cu CD-slit Model

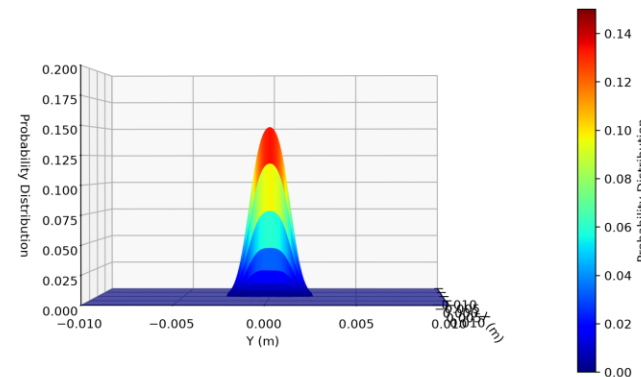
| Parameters         | Values  |
|--------------------|---------|
| Gas Flow Rate      | 3SLM    |
| Gas                | N2      |
| Nozzle Geometry    | CD-slit |
| Standoff Distance  | 10 mm   |
| Substrate Angle    | Flat    |
| Injection Particle | Cu      |



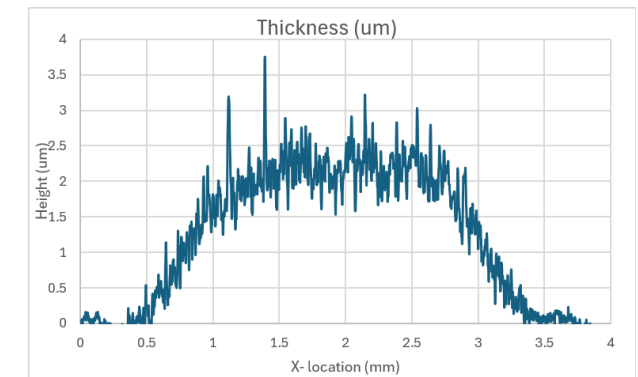
Particle locations Cu CD-slit



10s Cu CD-slit Birds Eye View



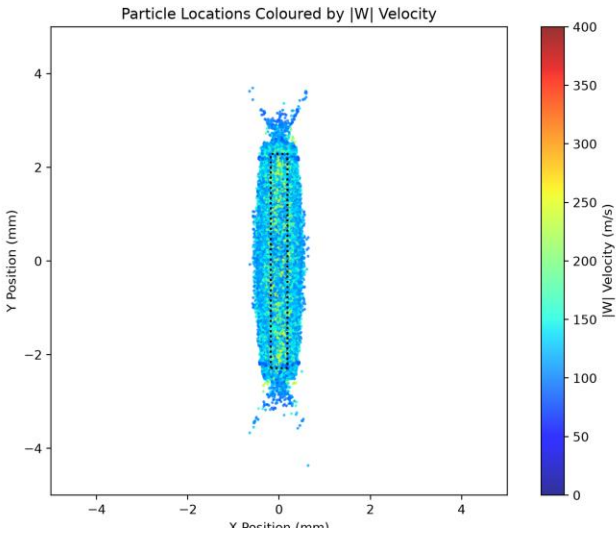
Cu CD-slit KDE Side Profile



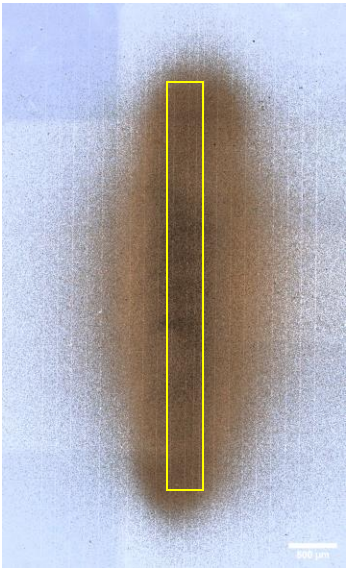
Cu CD-slit Cross-section

# Validating Cu CB-slit Model

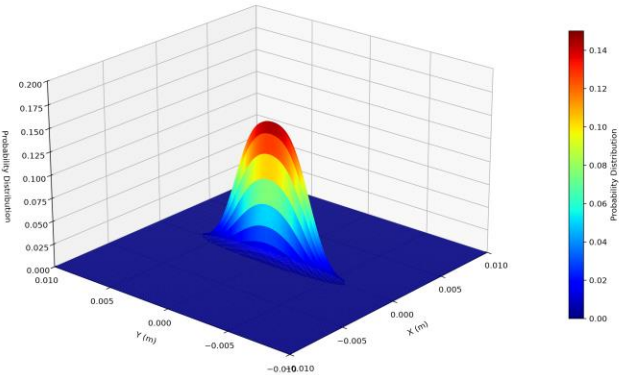
| Parameters         | Values  |
|--------------------|---------|
| Gas Flow Rate      | 3SLM    |
| Gas                | N2      |
| Nozzle Geometry    | CB-slit |
| Standoff Distance  | 10 mm   |
| Substrate Angle    | Flat    |
| Injection Particle | Cu      |



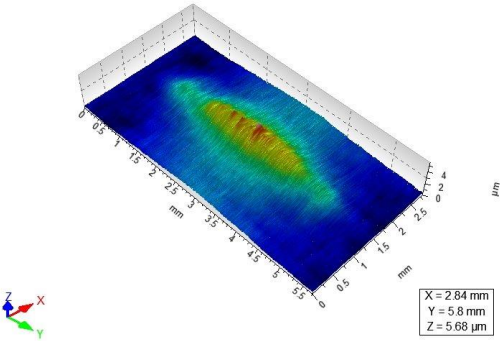
Particle Scatter of Cu CB-slit



10s Cu CB-slit Scatter



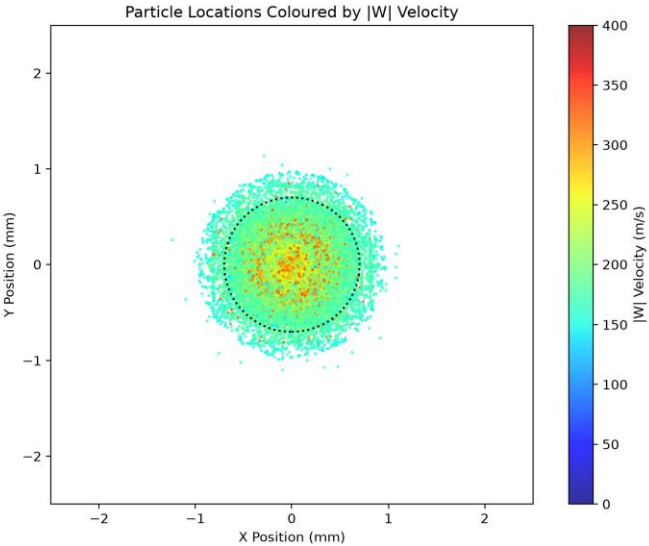
KDE of Cu CB-slit



10s Cu CB-slit 3D Profile

# Validating Cu CD-Round Model

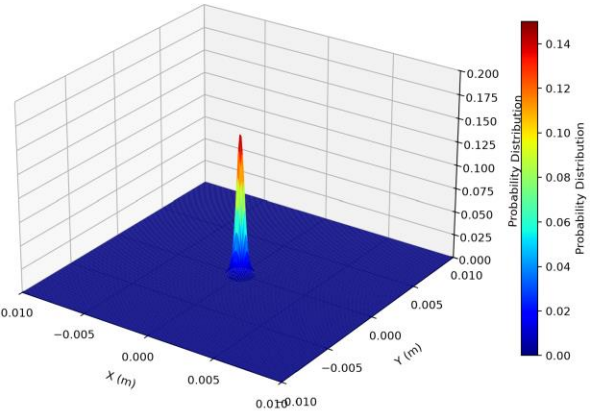
| Parameters         | Values   |
|--------------------|----------|
| Gas Flow Rate      | 3SLM     |
| Gas                | N2       |
| Nozzle Geometry    | CD-Round |
| Standoff Distance  | 10 mm    |
| Substrate Angle    | Flat     |
| Injection Particle | Cu       |



Particle Scatter of Cu CD-Round



10s Cu CD-Round Scatter



KDE of Cu CD-Round  
12

# A Modelling Workflow now Supports AD Repair

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## Completed:

- Refined Gas and Particle Modelling
- Implemented drag and heat-transfer correlations
- Automated ~500 CFD Cases
- Prepared outputs for FEM analysis and experimental comparison

## Next Steps:

- FEM Models to further investigate conditions for deposition
- Further validation through experimental comparisons

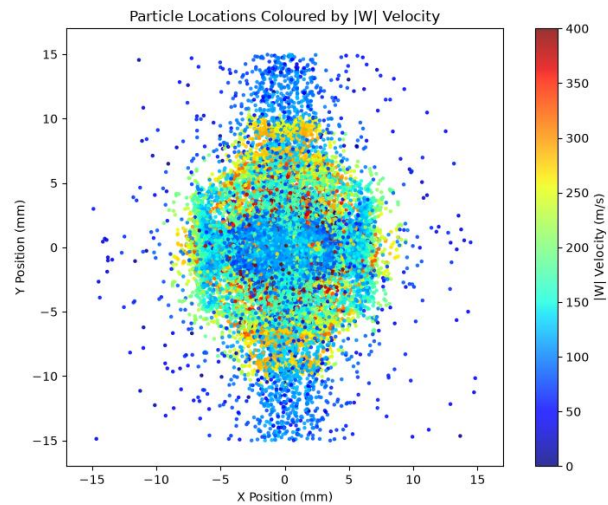
# Thank You



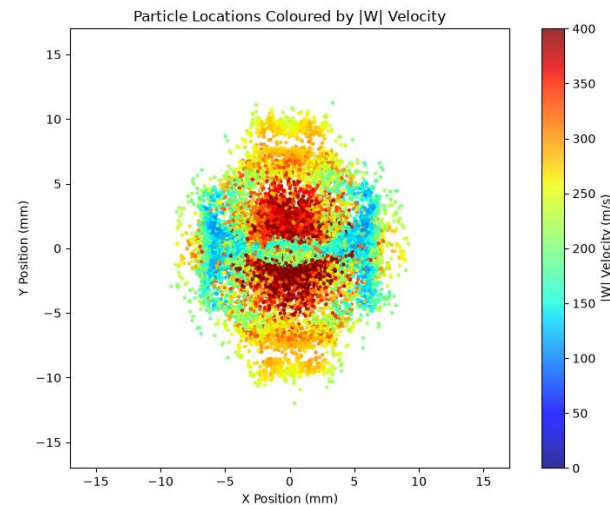
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# Custom Models improved Particle Tracking

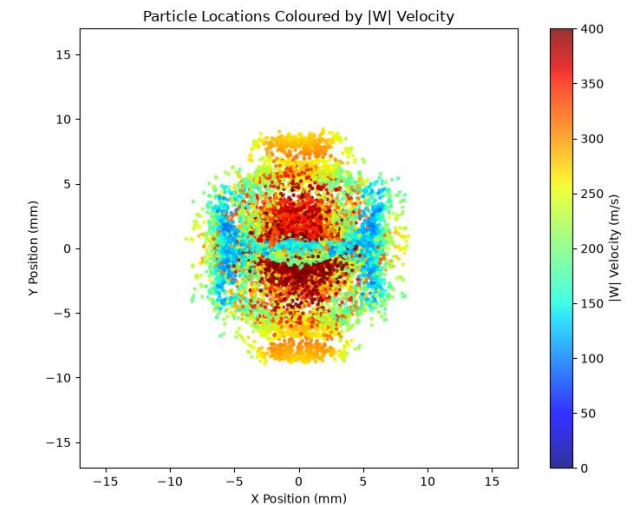
- Implemented Singh et al. Generalized spherical drag model
- 7.6% relative error reported compared 12.1 - 15.8 % reported in established models



Loth



Crowe



Singh

# Meshing

## Settings

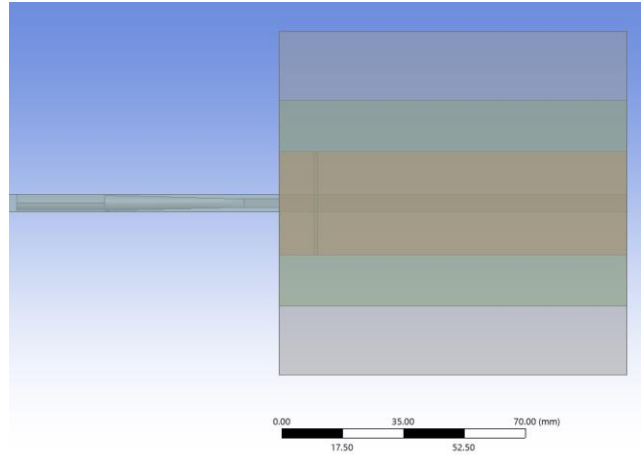
ANSYS Fluent Meshing

Refinement using BOIs

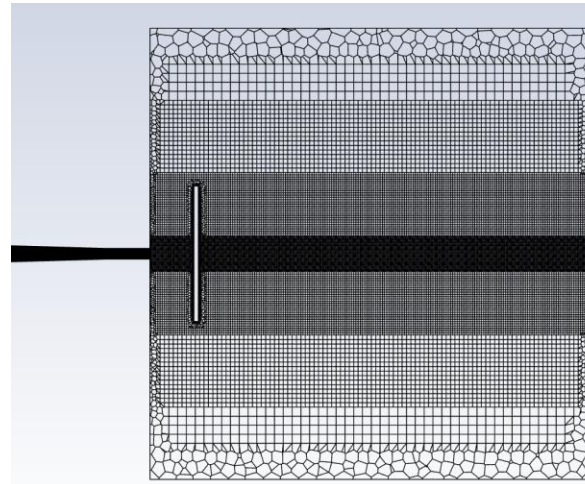
Only Fluid + Void Regions

Poly-Hexcore Volume Mesh

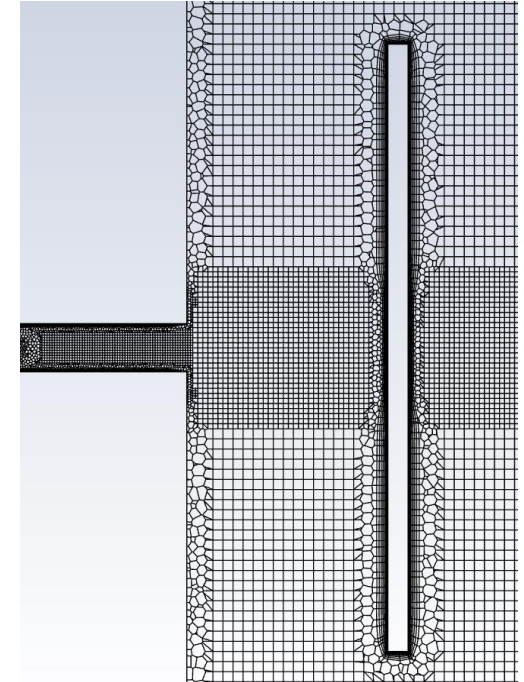
Target 3-6 million cells



Original Geometry Side Profile



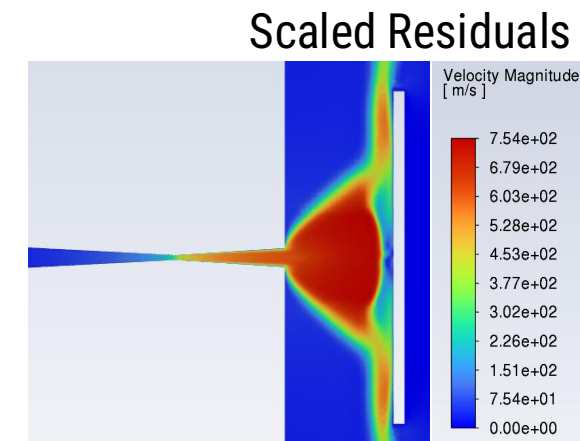
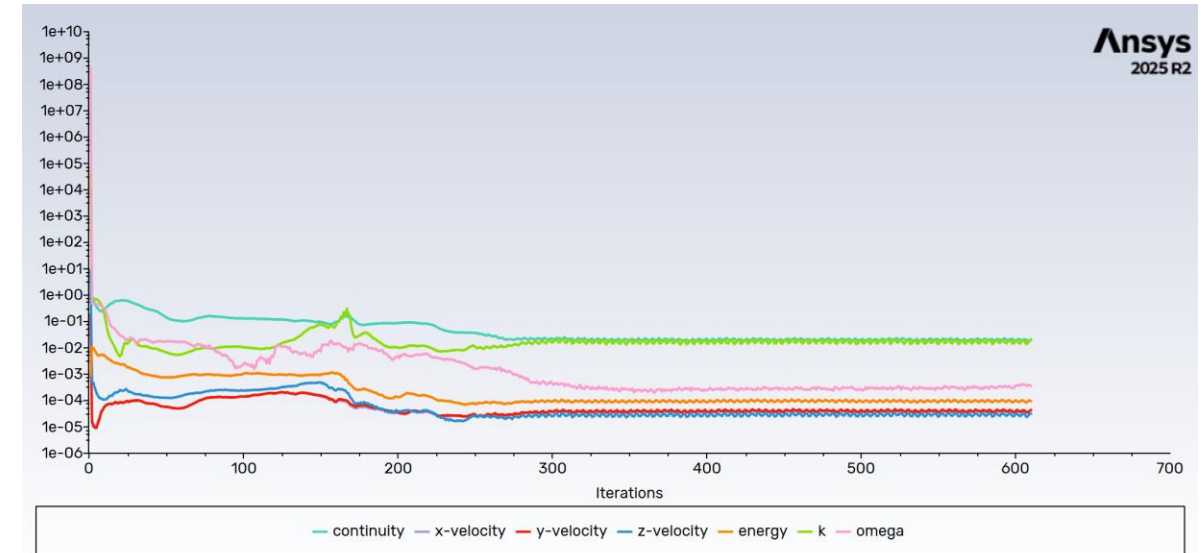
Mesh Side Profile



Closeup Mesh Side Profile

# Solving the Gas Flow Model

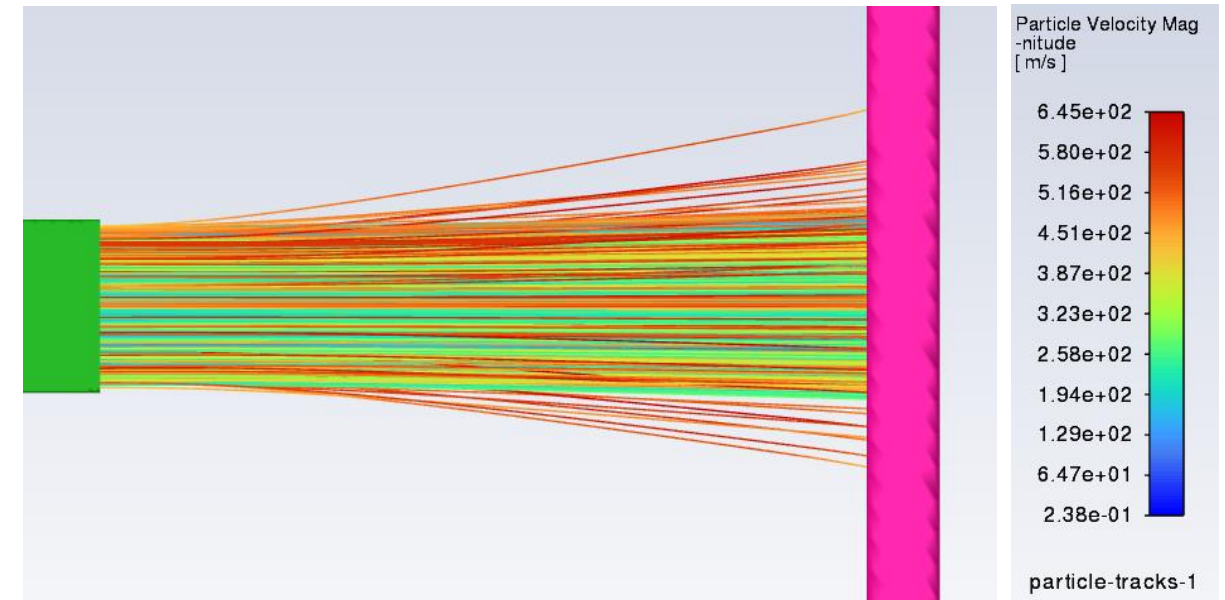
| Settings                            |                             |
|-------------------------------------|-----------------------------|
| Energy                              | On                          |
| Turbulence Model                    | SST k-omega                 |
| Gas Settings                        | Nitrogen w/ ideal gas model |
| Operating Conditions                | 133 Pa                      |
| Inlet Mass Flow Rate                | 10SLM --> 0.000193 kg/s     |
| Pressure Velocity Coupling          | Coupled                     |
| Pressure Spatial Discretization     | Standard                    |
| Momentum, and Energy Discretization | Second order upwind         |



Gas flow  
contour

# Applying DPM Model

| Settings                                                    |                            |
|-------------------------------------------------------------|----------------------------|
| Multiphase Model                                            | Eulerian-Langragian (DPM)  |
| Interaction with Continuous Phase                           | Enabled                    |
| Two way coupling for turbulence and Pressure Gradient Force | Enabled                    |
| Injection initial velocity                                  | 4m/s                       |
| Drag Law                                                    | Loth, Crowe, Singh         |
| Heat Transfer Law                                           | Nusselt                    |
| Particle Size Distribution                                  | Rosin-Rammler Distribution |
| Particle Injection                                          | Al2O3                      |



DPM Particle Tracks